Zero- Flow Engine

Beginner Setup Guide — Laptop / CPU Mode (no NVIDIA GPU)

Windows 11 · CPU mode, no graphics card needed · github.com/zerocold7/zero-flow-app

> **NOTE** **Have an NVIDIA, AMD or Intel graphics card?** Follow the Install Guide instead — it covers every kind of PC, and the Hardware Guide explains what to expect from yours.

This guide takes you from a fresh Windows 11 laptop to a working Zero- Flow setup, **one step at a time** — written for someone who has never installed something like this before. It uses **CPU mode**, so **you do not need an NVIDIA graphics card**.

Zero- Flow does two things, entirely on your own machine:

- **Dictation** — press a key, speak English or Arabic, press it again, and your words are typed wherever your cursor is.
- **Reading aloud** — highlight any text, press `F4`, and a natural voice reads it to you.

Just follow the steps in order. Every command is copy-paste.

What you need before you start

- Your laptop, signed in to Windows 11.
- An internet connection — you'll download about **9–11 GB** total (one time only).
- About **30–45 minutes** (most of it is just waiting for downloads).
- Roughly **13 GB** of free disk space.
- Your **built-in microphone** and speakers (no extra hardware needed).

> **TIP** **No NVIDIA GPU? That's completely fine.** This guide sets up **CPU mode**. It works exactly the same; your words just take about **1–4 seconds** to appear after you stop recording, instead of being instant.

Step 1 — Open PowerShell

PowerShell is the window where you'll type the setup commands.

1. Click the **Start** button (Windows logo, bottom-left).
2. Type the word **PowerShell**.
3. Click **Windows PowerShell** in the results. A dark window opens.

You'll type (or paste) one command at a time and press **Enter** after each. To paste a copied command, **right-click** inside the window.

Step 2 — Install the two required programs

You need two free programs: **Python** (runs the engine) and **Git** (downloads the project). The AI that polishes and translates is **built into the engine** — you install it in Step 7, with no extra program. Paste these two lines into PowerShell, pressing **Enter** after each:

```
winget install Python.Python.3.12
winget install Git.Git
```

When they finish, **close PowerShell and open it again** (Step 1) so it notices the new programs. Then check they both installed:

```
python --version
git --version
```

Each line should print a version number (for example `Python 3.12.x`).

> **TIP** **"winget is not recognized"?** Open the **Microsoft Store**, search for **App Installer**, and click **Update**. Then close and reopen PowerShell and try again.

> **TIP** **Typing `python` opens the Microsoft Store?** Go to **Settings → Apps → Advanced app settings → App execution aliases** and turn **OFF** the two entries named "python". (Or simply type `py` instead of `python` everywhere in this guide.)

Step 3 — Turn on the microphone (do not skip!)

Windows hides the microphone from desktop apps by default. If you skip this, the engine will record **silence** and type nothing.

1. Open **Settings** (Start menu → the gear icon).
2. Go to **Privacy & security** → **Microphone**.
3. Turn **ON** "Microphone access".
4. Turn **ON** "Let desktop apps access your microphone".

Step 4 — Download the project

> **IMPORTANT** **Avoid OneDrive.** New laptops often sync your **Desktop** and **Documents** folders to OneDrive, which can break file paths with confusing red errors during install. Put the project on the **`C:\` drive** directly instead.

Back in PowerShell, run these lines. They put the project in a folder at the root of your C: drive (`C:\zero-flow-app`):

```
cd C:\
git clone https://github.com/zerocold7/zero-flow-app.git
cd zero-flow-app
```

Step 5 — Laptop-friendly settings

Create your settings file and open it:

```
Copy-Item .env.example .env
notepad .env
```

Find these four lines and change them so they read exactly like this, then save (Ctrl+S) and close:

```
WHISPER_MODEL_NAME="small"
WHISPER_DEVICE="cpu"
LLAMA_VARIANT="cpu"
READER_DEVICE="cpu"
```

TIP **Why:** the default speech model (`large-v3`) is built for a big graphics card — on a laptop processor it takes longer than you took to speak. `small` for speech fits comfortably in a 16 GB laptop next to the built-in AI, and the three `cpu` lines tell the engine — the speech model, the AI and the voice — not to look for a graphics card.

TIP **Shortcut:** instead of editing by hand, run `python check_hardware.py --apply`. It looks at your laptop and writes the right lines for you — and if yours has only 8 GB of memory, it picks even lighter settings: the `base` speech model and the smaller AI model, Gemma 4 E2B.

Step 6 — Install the engine's Python parts

These commands create a private folder named **venv** and download the needed libraries into it. Run them **in this order** (the last one takes several minutes — wait until the prompt returns):

```
python -m venv venv
.\venv\Scripts\python.exe -m pip install --upgrade pip
$parts = Get-Content requirements.txt |
    Where-Object { $_ -match '^[A-Za-z]' -and $_ -notmatch '^nvidia-' }
.\venv\Scripts\python.exe -m pip install $parts -c constraints.txt
```

The last three lines read the list of parts from the project's own file and skip the two lines that start with `nvidia-`: they are only for NVIDIA graphics cards and would waste ~1.2 GB. It still installs the reading voice (`kokoro`). Nothing in the project's files is changed, so later updates (`git pull`) go through cleanly.

Step 7 — Install the built-in AI

The AI behind Polish and Translate runs inside the engine — the engine starts it and stops it by itself. Install its program once, with this line:

```
.\venv\Scripts\python.exe tools\get_llama.py
```

It picks the right version for your laptop (here: the small `cpu` one, about 19 MB), downloads it from the official llama.cpp project on GitHub, checks the download is genuine, and puts it in the `runtime` folder inside the project. Wait until it prints `installed:`.

The **AI model** itself — Google's **Gemma 4 E4B**, about **5.15 GB** — downloads by itself the first time you start the engine (Step 8). No account and no extra command are needed. On a laptop with only 8 GB of memory, `check_hardware.py --apply` (Step 5) picks the smaller **Gemma 4 E2B** (3.35 GB) instead; its Arabic has not been tested yet.

Step 8 — Start it for the first time

1. Open **File Explorer** and go to `C:\zero-flow-app`.
2. Double-click `Launch_Zero.bat`.

> **IMPORTANT** **If Windows shows a blue "Windows protected your PC" box**, click **More info**, then **Run anyway**. This is just Windows being cautious about a new file — it's your own project file.

A console window opens. The **first** launch downloads the speech model (~500 MB), the reading voice (~350 MB) and the AI model (~5.15 GB) — let it finish; on a slow connection this can take a while. Later launches load everything from your own disk and don't need the internet at all.

Dictation is ready when the window shows a box titled `Z E R O - F L O W E N G I N E` with the list of keys under it. **Reading aloud is ready** a little later, when this line appears:

```
[TTS] Voice model ready.
```

Leave that window open (you can minimize it). **Closing the window stops the engine.**

Step 9 — Test that it works

> **IMPORTANT** **Laptop Fn-key trap (important!).** On most laptops, **F5–F10 are media keys** (volume/brightness) by default, so pressing `F5` alone won't reach the engine. Fix it once: press `Fn + Esc` to turn on **Fn-Lock** (this is the Lenovo/IdeaPad shortcut) — now the F-keys work with a single press. Or hold `Fn` together with the key every time (e.g. `Fn + F7`). If `Fn + Esc` doesn't work, look for a small padlock icon on the `Esc` key, or check your laptop's manual.

Dictation:

1. Open **Notepad** and click inside it so the cursor is blinking.
2. Press **F5** once (you'll hear a short beep), say "hello, this is a test", then press **F5** again.
3. After a second or two, your words appear in Notepad.
4. Try **F7** and speak Arabic the same way.

> **TIP** **Tap, don't hold.** Each key starts recording on the first press and stops on the second. Holding it down makes it start and stop over and over.

Reading aloud:

1. Highlight a sentence in any window — the text you just dictated works.
2. Press **F4**. The voice reads it out. Press **Esc** to stop it early.

If nothing happens, see **Troubleshooting** below.

Step 10 — Hotkey reference

Press the key, speak, press it again. The key **forces** the language, so it's never misheard.

Key	What it does
`F5`	Dictate **English — raw** (exactly as spoken)
`F6`	Dictate **English — polished** (AI cleans grammar/fillers)
`F7`	Dictate **Arabic — raw**
`F8`	Dictate **Arabic — polished**
`F9`	**Translate** English → Arabic

Key	What it does
`F10`	**Translate** Arabic → English
`Shift + F1`	Tidy up the learned-vocabulary file (the old list is kept as a backup)
`Shift + F2`	Clear the debug log and dictation history
`Shift + F3`	Rewrite and fix the current line
`F4`	**Read the highlighted text aloud** (see Step 12)
`Esc`	Cancel the recording, or silence the reading

Spoken commands work inside any dictation:

- Start with **"new line"** to begin a new line, **"bullet"** for a • bullet point, or **"format code"** to wrap the text as code.
- End with **"period"**, **"comma"** or **"question mark"** to type the real mark.
- In Arabic: every command has an Arabic equivalent — they are listed in the Arabic guide (SETUP-CPU-LAPTOP.ar.md) and in CONFIGURATION.md.

Step 11 — Everyday use: starting and stopping

- **Start:** double-click `Launch_Zero.bat` and minimize the window. (`Launch_Flow.bat` starts dictation only, without `F4`.)
- **Stop:** close that console window (click the **X**), or right-click the tray icon and choose **Exit Engine**.
- **Find the tray icon:** click the small **^** arrow next to the clock; the Zero- Flow icon lives there.
- **The AI** starts and stops with the engine, so there is nothing else to start or keep running.

> **TIP** **Run it invisibly:** double-click `Launch_Zero_Silent.vbs` to start with no window at all. To stop it, right-click the tray icon and choose **Exit Engine**. Stick with the visible `.bat` until you're comfortable — its window shows you what the engine is doing.

Step 12 — Reading text aloud (`F4`)

Highlight any text — a web page, a PDF, a chat message — and press `F4`. A local voice reads it out. Press `Esc` to stop it; to read something else, press `Esc`, highlight the new text and press `F4`.

`F4` comes with `Launch_Zero.bat`, which this guide uses. (`Launch_Flow.bat` is dictation only.) The reader stays quiet while you are dictating, so it never talks over your microphone.

Right-click the tray icon to switch voice, to pause the reader (**Suspend Reader**), or to turn on **Smart LLM Cleaning**, which asks the local AI to strip page clutter (menus, cookie notices) before reading.

> **NOTE** **On a laptop with no NVIDIA card, expect a short pause before the voice starts.** The voice model loads in the background when the engine starts; wait for `Voice model ready.` before your first `F4`. After that the voice starts once the first sentence is ready and keeps going smoothly while the rest is prepared. Reading a paragraph at a time rather than a whole page keeps that first pause short.

Troubleshooting

If this happens…	Do this
Nothing is typed when I speak	Check microphone permissions (Step 3). Press the key **once** to start and **once more** to stop — don't hold it. Look at the console window for a red error message.
The console says "Could not open the microphone"	The microphone is off, unplugged, or blocked — go through Step 3 again, then restart the engine.
`python is not recognized`	Close and reopen PowerShell. If it still fails, reinstall Python and tick "Add to PATH", or type `py` instead of `python`.
`winget is not recognized`	Update **App Installer** from the Microsoft Store, then reopen PowerShell.
It types in the wrong language	Use the matching key: F5/F6 for English, F7/F8 for Arabic. The key decides the language.
Hotkeys do nothing in one specific app	Right-click `Launch_Zero.bat` and choose **Run as administrator**.
Polish / Translate types my words unchanged	Look at the console when the engine starts: if it says the built-in AI is not installed, do Step 7 and restart the engine. Otherwise open `flow_debug.log` and look for `Built-in AI unavailable` — the reason follows, and `builtin_ai.log` in `C:\zero-flow-app` has the AI's own messages.
`F4` **does nothing**	Make sure the text is **highlighted** first, and that the window has shown `Voice model ready.` The reader won't speak while a dictation is recording. Check you started `Launch_Zero.bat`, not `Launch_Flow.bat`.
Everything feels too slow	In `.env` set `WHISPER_MODEL_NAME="base"`, save, and restart the engine.
SmartScreen blocks the launcher	Click **More info**, then **Run anyway**.
Pressing F5–F10 does nothing at all	Your laptop's F-keys are in media mode. Press `Fn + Esc` to enable Fn-Lock (Step 9), or hold `Fn` together with the key.
The laptop gets hot / fans get loud	Use the raw modes `F5`/`F7` (they skip the AI). Confirm `.env` has `small` (not `medium`). Drop to `base`, or to the smaller AI model, for more cooling. See below.
Something else went wrong	Open `flow_debug.log` (dictation) or `reader_debug.log` (reading) in `C:\zero-flow-app` with Notepad — the last lines say what happened.

Keeping your laptop cool & fast

On a CPU laptop, **bigger models = more heat.** The biggest heat source is the AI model (the built-in Gemma), which works your CPU hard for a few seconds whenever you **polish** or **translate**. To stay cool:

- **For the least heat, use the raw modes `F5` (English) and `F7` (Arabic).** They type exactly what you say **without** running the AI, so the CPU barely warms up. Only use `F6`/`F8` (polish) and `F9`/`F10` (translate) when you actually need them — those are what run the AI model.
- **The heat is momentary.** The engine only works the CPU while it's processing (the few seconds after you stop a recording, or while it prepares speech). The rest of the time it sits idle and cool.
- **Still warm or slow?** Set `WHISPER_MODEL_NAME="base"` in `.env` (faster and cooler), then restart.
- **Keep `.env` light:** `small` for speech. Avoid `medium` on a laptop — it's the usual cause of fan noise and heat. For a lighter AI, switch to Gemma 4 E2B (see the table below).
- **Airflow:** use the laptop on a hard surface (not a bed or cushion) so the vents aren't blocked.

Making it faster or more accurate

After changing any of these in `.env`, **save the file and restart the engine** (close the console and run the launcher again).

Speech model (`WHISPER_MODEL_NAME`):

Value	Trade-off
`tiny` / `base`	Fastest, lowest accuracy. Good if "small" feels slow.
`small`	Recommended balance for a laptop (this guide's default).
`medium`	Better Arabic accuracy, noticeably slower on CPU.

AI model (two lines, `AI_BUILTIN_REPO` and `AI_BUILTIN_FILE` — the new one downloads by itself on the next start):

Lines in `.env`	Trade-off
`AI_BUILTIN_REPO="google/gemma-4-E4B-it-qat-q4_0-gguf"` `AI_BUILTIN_FILE="gemma-4-E4B_q4_0-it.gguf"`	Recommended — the default; good Arabic + English.
`AI_BUILTIN_REPO="google/gemma-4-E2B-it-qat-q4_0-gguf"` `AI_BUILTIN_FILE="gemma-4-E2B_q4_0-it.gguf"`	Smaller and lighter, for 8 GB laptops. Its Arabic has no

Prefer Ollama or another AI service? See §6 of CONFIGURATION.md.

PRIVACY **Privacy:** everything in this guide runs **locally on your machine** — your voice and text are never uploaded. For deeper customization (other languages, other AI providers, prompts), see **CONFIGURATION.md**.